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Department of Mathematics

VECTOR CALCULUS, LAPLACE TRANSFORM & NUMERICAL METHODS (MA221TA)

UNIT-III- LAPLACE TRANSFORM

TUTORIAL SHEET-1

Find the following:

1. $L\left[\sqrt{t}-\frac{1}{\sqrt{t}}\right]$ =_____. Ans: $\sqrt{\pi}\left(\frac{1}{2s^{\frac{3}{2}}}-\frac{1}{s^{\frac{1}{2}}}\right)$

2. $L[e^{-t}2^t]$ = _____. Ans: $\frac{1}{s+1-\log 2}$

3. $L\left[\sin 4t+\cos t\cos 2t\right]$ = _____. Ans: $\frac{4}{s^2+16}+\frac{1}{2}\left(\frac{s}{s^2+9}+\frac{s}{s^2+1}\right)$

4. $L\left[a^t\right]$ =_____. Ans: $\frac{1}{s-\log a}$

5. Evaluate the Laplace transform of the following signals:

(i) $f(t) = e^{-3t}(t+1)^2$ Ans: $\frac{2}{(s+3)^3}+\frac{2}{(s+3)^2}+\frac{1}{s+3}$

(ii) $f(t) = e^{2t}\sin 3t$ Ans: $\frac{3}{(s-2)^2+9}$

(iii) $f(t) = e^{at}\cosh bt$ Ans: $\frac{s-2}{(s-2)^2-b^2}$

(iv) $f(t) = e^{\frac{3}{2}t}\cos^3 t$ Ans: $\frac{1}{4}\left[\frac{s-\frac{3}{2}}{\left(s-\frac{3}{2}\right)^2+9}+\frac{3\left(s-\frac{3}{2}\right)}{\left(s-\frac{3}{2}\right)^2+1}\right]$

(v) $f(t) = t(1-\sqrt{t}e^t)^3$ Ans: $\frac{1}{s^2}-\frac{15\sqrt{\pi}}{8(s-3)^{\frac{7}{2}}}-\frac{9\sqrt{\pi}}{(s-1)^{\frac{5}{2}}}+\frac{6}{(s-2)^3}$

(vi) $f(t) = e^{at}\cos(bt+a)$ Ans: $\cos a\frac{s-a}{(s-a)^2+b^2}-\sin a\frac{b}{(s-a)^2+b^2}$

(vii) $f(t) = \sin t\sin 2t\sin 3t+e^{-3t}\cosh^2 2t$

Ans: $\frac{1}{4}\left[\frac{2}{s^2+4}+\frac{4}{s^2+16}-\frac{6}{s^2+36}+\frac{1}{s-1}+\frac{1}{s+7}+\frac{2}{s+3}\right]$

(viii) f(t) = $\begin{cases} t^2, & 0\le t<1 \\ 2-t, & 1\le t<2 \\ 7, & t\ge 2 \end{cases}$. Also draw the graph of f(t).

Ans: $\frac{e^{-2s}}{s^2} - \frac{3e^{-s}}{s^2} + \frac{7e^{-2s}}{s} - \frac{2e^{-s}}{s^3} + \frac{2}{s^3}$

(ix) f(t) = $\begin{cases} e^t, & 0 < t < 1 \\ 0, & t > 1 \end{cases}$ Ans: $\frac{1}{1-s}[1 + e^{1-s}]$

(x) f(t)=$\sin^5 t$ Ans: $\frac{5}{16(s^2+25)} - \frac{15}{16(s^2+9)} + \frac{5}{8(s^2+1)}$

(xi) f(t) = $\begin{cases} (t-a)^3, & t > a \\ 0, & t < a \end{cases}$ Ans: $\frac{6}{s^4}e^{-as}$

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UNIT-III- LAPLACE TRANSFORM

TUTORIAL SHEET-2

1. If $L\{f(t) = F(s)$ then $L\{f(3t)\} =$ _____. Ans: $\frac{1}{3}F\left(\frac{s}{3}\right)$

2. $s^3F(s) - s^2f(0) - sf'(0) - f''(0) =$ ____. Ans: $L\{f^3(t)\}$

3. Evaluate $\int_0^\infty e^{-t}t^2 \cos 2t\, dt$. Ans: $-\frac{22}{125}$

4. Evaluate L $\left\{\int_0^t \frac{\sin 4t}{e^{-t}\,t}dt\right\}$. Ans: $\frac{1}{s}\cot^{-1}\left(\frac{s-1}{4}\right)$

5. Find the Laplace transform of $\left(\sqrt{t} - \frac{1}{\sqrt{t}}\right)^3$. Ans: $\frac{\sqrt{\pi}}{s}\left(\frac{3}{4s^2} - \frac{3}{2s} + 3 + 2s\right)$

6. Find $L\{t^n + \sin at\}$, where n is a positive integer & $a > 0$. Ans: $\frac{n!}{s^{n+1}} + \frac{a}{s^2+a^2}$

7. Find $L\left\{\cos^3 2t + e^{-3t}(2\cos 5t - 3\sin 5t)\right\}$.

Ans: $\frac{1}{4}\left[\frac{s}{s^2+36} + \frac{3s}{s^2+4}\right] + \frac{2(s+3)}{(s+3)^2+25} - \frac{15}{(s+3)^2+25}$

8. Given $L\{2\sqrt{t/\pi}\} = \frac{1}{s^{3/2}}$, show that $L\left\{\frac{1}{\sqrt{\pi t}}\right\} = \frac{1}{\sqrt{s}}$.

9. Prove that $e^{2t}\int_0^\infty (te^{-2t}\sin 3t\, dt) = \frac{12}{169}$.

10. Find $L\left(\frac{\cos 5t - \cos 6t}{t}\right)$. Ans: $\log\left(\sqrt{\frac{s^2+36}{s^2+25}}\right)$

11. Obtain $L\left(\int_0^t \frac{\sin^2 t}{t}dt\right)$. Ans: $\frac{1}{2s}\log\left(\frac{\sqrt{s^2}+4}{s}\right)$

12. Find the Laplace transform of $t^2e^t \sin 4t$. Ans: $\frac{8(3s^2-6s-13)}{(s^2-2s+17)^3}$

13. Find the Laplace transform of $t^2(e^{-2t} - \cos 2t + 4)$ Ans: $\frac{2}{(s+2)^3} + \frac{8}{s^3} - \frac{2s(s^2-12)}{(s^2+4)^3}$

14. Prove that $\int_0^\infty \frac{e^{-t}-e^{-3t}}{t}\, dt = log3$

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UNIT-III- LAPLACE TRANSFORM

TUTORIAL SHEET-3

1. A rectangular wave $f(t)$ of period $2a$, $a>0$ is defined by

$$f(t)=\begin{cases} E, & 0\le t\le a \\ -E, & a<t\le 2a \end{cases}$$

Show that $L\{f(t)\}=\frac{E}{s}tanh\left(\frac{as}{2}\right)$. Also draw the graph of the wave function.

2. A periodic function of period $2\pi/\omega$ is defined by

$f(t)=\begin{cases} E\sin\omega t, & 0\le t<\pi/\omega \\ 0, & \pi/\omega\le t\le 2\pi/\omega \end{cases}$, where E & ω are positive constants.

Show that $L\{f(t)\}=\frac{E\omega}{(s^2+\omega^2)(1-e^{-\pi s/\omega})}$. Also draw the graph of the function.

3. Find the Laplace transform of function f(t)= $t\,u(t-4)-t^2\delta(t-2)$

Ans: $\frac{e^{-4s}1}{s^2}(1+4s)-4e^{-2s}$

4. Evaluate $\int_0^\infty t^m(\log t)^n\,\delta(t-3)dt$. Ans:$3^m\log 3^n$

5. Express $f(t)=\begin{cases}\sin t, & 0<t\le\pi/2 \\ \cos t, & t>\pi/2\end{cases}$ in terms of the unit step function and hence find its Laplace Transform. Ans: $\frac{1}{s^2+1}\left\{1-e^{-\frac{\pi s}{2}}(s+1)\right\}$

6. Express $f(t)=\begin{cases}2, & 0<t\le\pi \\ 0, & \pi<t<2\pi \\ \sin t, & t>2\pi\end{cases}$ in terms of the unit step function and hence find its Laplace Transform.

7. Express $f(t)=\begin{cases}t^2, & 0<t\le 2 \\ 4t, & t>2\end{cases}$ in terms of the unit step function and hence find its Laplace Transform. Ans: $\frac{2}{s^3}+e^{-2s}\left(-\frac{2}{s^3}+\frac{4}{s}\right)$

8. Find Laplace of periodic function $f(t)$ of period $2a, a>0$ defined by

$f(t)=\begin{cases} t, & 0 \le t \le a \\ -t+2a, & a < t \le 2a \end{cases}$. Also draw the graph of the function. Ans: $\frac{1}{s^2}\tanh\frac{as}{2}$